

PHK Spike Episodes 2022–2026

DEN's provincial PHK flow data, cross-checked against the reported causal narratives — do the media-attributed causes hold up against the actual pace of layoffs?

1. Purpose and Data Basis

This note consolidates two things: (a) DEN's own PHK flow data — province × month, 2022–2026, 38 provinces × 60 months, sourced from `phk_master.csv` — used to identify statistically extreme layoff months, and (b) two rounds of media/event research testing whether the causal narratives commonly attached to those spikes actually match the pace of the underlying data.

The data was screened by computing, for each province, its own historical mean and standard deviation of monthly PHK flow, then flagging any month where the flow exceeded 2× or 4× that province's SD. This per-province approach (rather than one pooled national threshold) avoids mistaking a normal month in a large province for an anomaly purely due to scale.

1.1 National Overview — Where the Extreme Months Sit

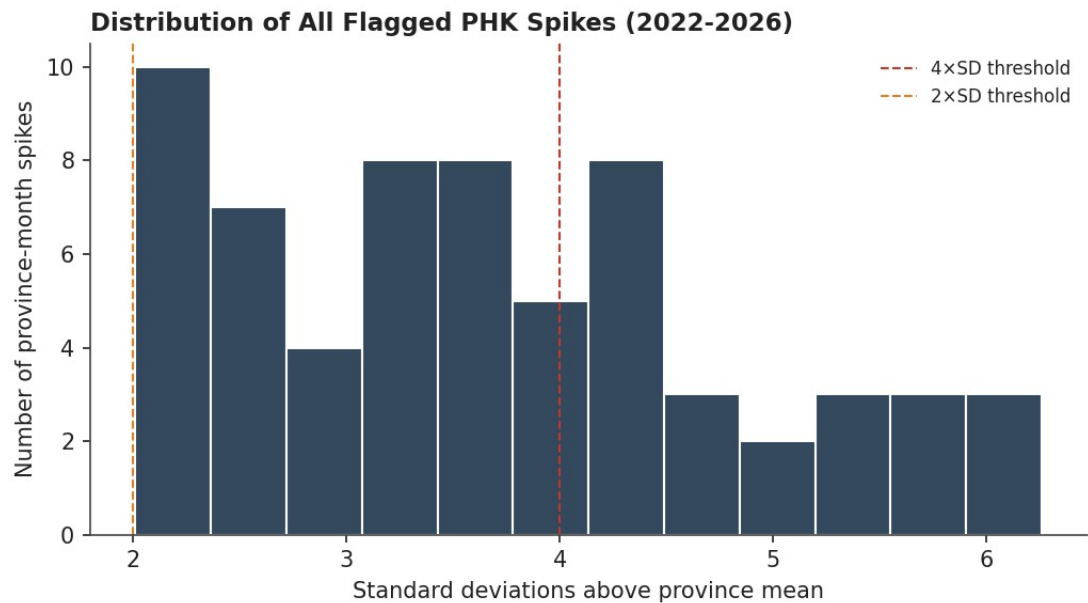


Figure 1. Distribution of all flagged province-month PHK spikes by SD multiple. Dashed lines mark the 2× and 4× SD thresholds used for tiering.

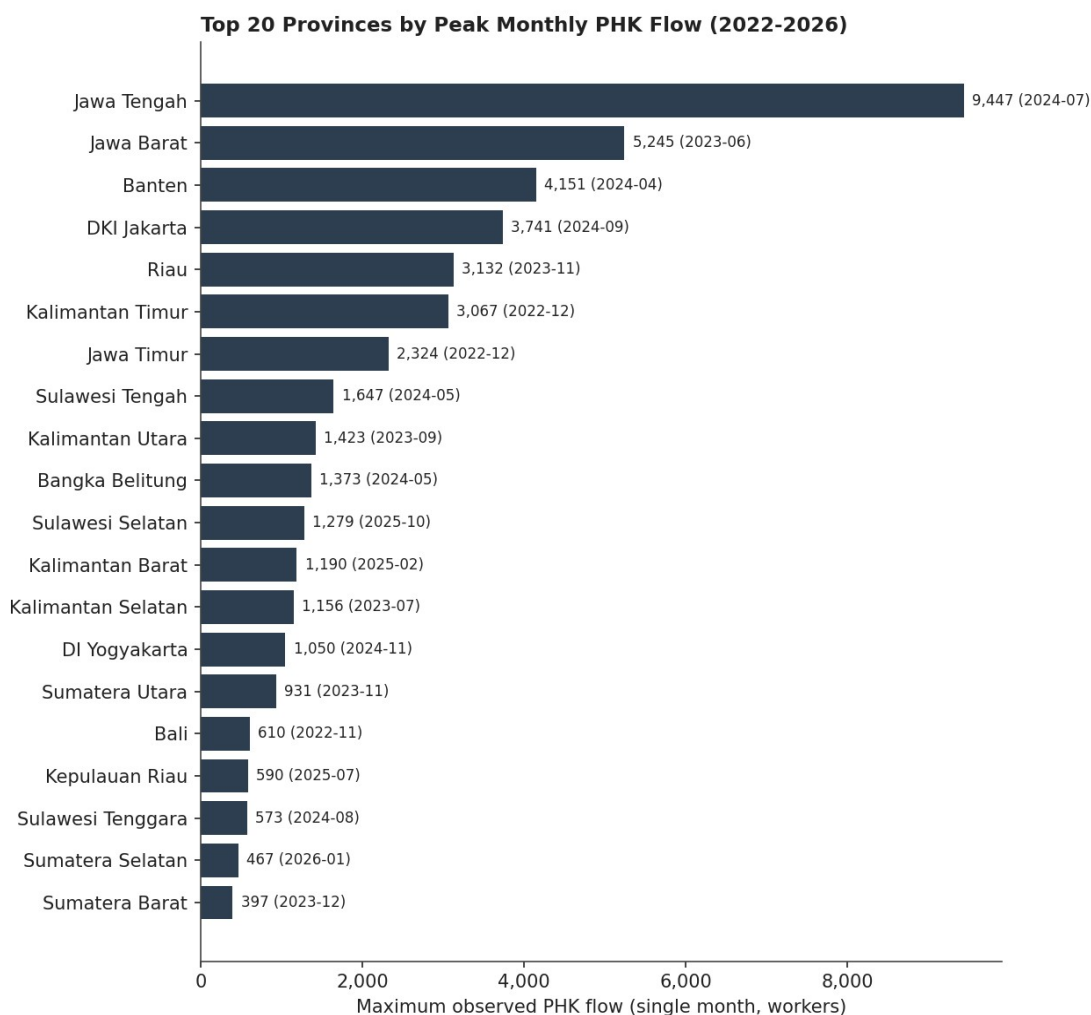


Figure 2. The 20 provinces with the single highest monthly PHK flow ever recorded, 2022–2026, with the month it occurred.

1.2 Full Province-Level Outlier Summary

Below is the complete per-province statistical profile: each province's peak month, its own historical mean and SD, and how many months crossed the $2\times SD$ and $4\times SD$ thresholds. Sorted by peak magnitude.

Province	Max Value	Max Month	Mean	SD	# > 2SD	# > 4SD
Jawa Tengah	9,447	2024-07	887	1,990	2	1
Jawa Barat	5,245	2023-06	1,329	1,116	4	0
Banten	4,151	2024-04	852	1,041	2	0
DKI Jakarta	3,741	2024-09	642	965	4	0
Riau	3,132	2023-11	207	483	1	1
Kalimantan Timur	3,067	2022-12	277	479	1	1
Jawa Timur	2,324	2022-12	431	519	3	0
Sulawesi Tengah	1,647	2024-05	163	356	2	1
Kalimantan Utara	1,423	2023-09	85	246	1	1
Bangka Belitung	1,373	2024-05	58	210	1	1
Sulawesi Selatan	1,279	2025-10	161	248	2	1

Province	Max Value	Max Month	Mean	SD	# > 2SD	# > 4SD
Kalimantan Barat	1,190	2025-02	125	270	3	0
Kalimantan Selatan	1,156	2023-07	183	252	3	0
DI Yogyakarta	1,050	2024-11	125	191	2	1
Sumatera Utara	931	2023-11	131	183	2	1
Bali	610	2022-11	37	95	1	1

Table 1. Top 16 provinces by peak monthly PHK flow (full 34-province table available in `phk_outlier_summary.csv`).

2. Batch-by-Batch: Data, Narrative, and Pace Verification

Each spike cluster below was first researched for a plausible causal narrative (Round 1), then tested against actual monthly/quarterly PHK pace data (Round 2) to determine whether the cited cause coincided with a genuine acceleration in layoffs, or whether PHK was already elevated/trending before the event.

Batch 1 — November–December 2022

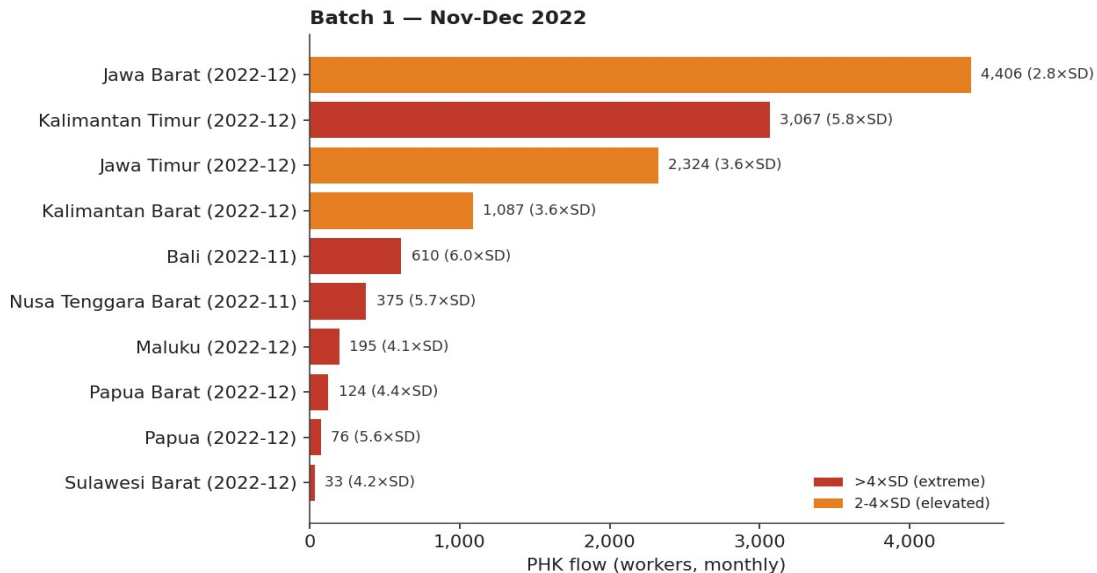


Figure 3. Batch 1 province-month spikes, sorted by magnitude. Red = $>4\times SD$ (extreme), orange = $2-4\times SD$ (elevated).

Province	Month	PHK Flow	SD Multiple	Tier
Kalimantan Timur	2022-12	3,067	5.82×	4SD+
Bali	2022-11	610	6.01×	4SD+
Jawa Timur	2022-12	2,324	3.65×	2-4SD
Jawa Barat	2022-12	4,406	2.76×	2-4SD
Nusa Tenggara Barat	2022-11	375	5.70×	4SD+
Kalimantan Barat	2022-12	1,087	3.56×	2-4SD
Maluku	2022-12	195	4.14×	4SD+
Papua Barat	2022-12	124	4.41×	4SD+
Papua	2022-12	76	5.58×	4SD+
Sulawesi Barat	2022-12	33	4.22×	4SD+

Original attribution (Round 1)

Textile/garment export collapse (global demand weakness, Russia-Ukraine war effects) reaching 70,000 PHK by November 2022, plus a wave of tech/startup layoffs (GoTo, Shopee, LinkAja, Tokocrypto, Ruangguru, and others).

Pace verification (Round 2)

- Kemnaker's official full-year 2022 total was only 25,114 — far below the ~79,316 garment-sector figure reported separately by Apindo on 16 November 2022, itself up 15,316 in October–November alone. Apindo explicitly dated the pressure to “sejak awal semester kedua tahun 2022” (since the start of H2 2022) — the Nov–Dec peak was the visible top of a five-month build, not a sudden event.

- Footwear layoffs (25,700 across 37 factories) were tied to a 45% order decline “sejak Juli-Oktober 2022” — again predating the Nov-Dec window.
- Major tech layoffs were spread across the whole year, not clustered at year-end: Pahamify (June), Zenius (May-August), Shopee Indonesia (September), Fabelio bankruptcy (October); only GoTo (18 Nov) and JD.ID (December) actually fell inside the window.
- Kalimantan Timur: no documented discrete Oct-Dec 2022 event found even after broadening the search window. Flagged as a likely data artifact.
- Bali, NTB, Papua, Sulawesi Barat, Maluku, Papua Barat: no documented events found — most plausibly small-sample statistical noise given tiny absolute magnitudes.

Verdict: Continuation, not new shock. Textile/tech distress was already well underway; the province-month spikes mark where it became statistically visible in the JKP-claims data, not where it began.

Batch 2 — June–September 2023

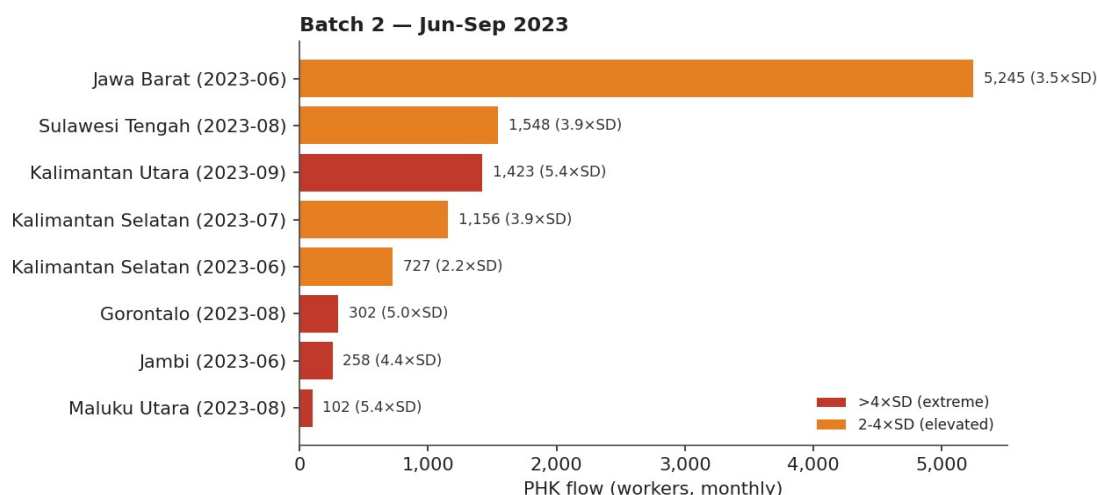


Figure 4. Batch 2 province-month spikes, sorted by magnitude.

Province	Month	PHK Flow	SD Multiple	Tier
Riau	2023-11	3,132	6.06×	4SD+
Jawa Barat	2023-06	5,245	3.51×	2-4SD
Jawa Barat	2023-03	3,809	2.22×	2-4SD
Kalimantan Utara	2023-09	1,423	5.44×	4SD+
Sulawesi Tengah	2023-08	1,548	3.90×	2-4SD
Kalimantan Selatan	2023-07	1,156	3.86×	2-4SD
Kalimantan Selatan	2023-06	727	2.16×	2-4SD
Gorontalo	2023-08	302	5.05×	4SD+
Jambi	2023-06	258	4.43×	4SD+

Original attribution (Round 1)

Coal price correction (HBA benchmark decline) affecting Riau, Kalimantan Selatan, Kalimantan Utara; continued textile crisis affecting Jawa Barat; a palm oil (CPO) price story for Jambi.

Pace verification (Round 2)

- HBA coal benchmark fell gradually all year (US\$305.21 in January to US\$117.38 in December), with the single steepest monthly drop in May 2023 (-22.28%) — before the Jun-Sep window.
- National monthly PHK cumulative data was roughly linear through H1 2023 (~4,400/month) and only modestly higher Jul-Sep (~5,000+/month) — the genuine acceleration point was actually October-November 2023 (~7,800/month).
- The oft-cited “358,809 total 2023 layoffs” figure is unreliable — roughly 5.5× Kemnaker's official full-year count of 64,855. Should be dropped or heavily caveated.
- Real named coal-sector layoffs were found in Kalimantan Utara (Nunukan) — PT Duta Tambang Rekayasa (203 workers, reserve exhaustion, May-June 2023) and Site Sebakis firms (~286 workers, coal quality, March-April 2023) — not in Riau or Kalimantan Selatan as the original attribution implied.
- Jambi's CPO reference price declined gradually through H1 2023 (Rp12,588/kg in March to Rp9,446/kg by June) — a slow erosion, not a mid-year cliff.

Verdict: No discrete trigger in the cited window. The commodity story is real but its timing doesn't match; the genuine acceleration point was two months later than attributed.

Batch 3 — July 2024 (Largest Single Spike in the Dataset)

This batch is a single province-month observation, so a table alone is sufficient here — no chart needed.

Province	Month	PHK Flow	SD Multiple	Tier
Jawa Tengah	2024-07	9,447	4.30×	4SD+

Original attribution (Round 1)

PT Sai Apparel bankruptcy in Semarang (8,000 workers, dated 8 July 2024) — the single largest documented event in the whole dataset.

Pace verification (Round 2)

- National monthly PHK for 2024 rose steadily and broadly all year (~3,330 in January to ~8,390 in May, 77,965 for the full year) — a broad-based climb, not isolated monthly spikes.
- The flagship Sai Apparel attribution needs correction: the viral “8,000 PHK, 8 July 2024” claim was disputed by the plant union (KSPN), FKSPN Jateng, and Disnakertrans Jateng. The actual PHK was 1,482 workers, executed 31 December 2023, tied to a relocation to Grobogan. The Semarang plant kept operating with ~2,500 workers into 2024. (CNBC Indonesia and Kemenperin separately list Sai Apparel among 2024 closures with ~8,000 affected — a genuine sourcing conflict, not a settled correction.)
- Jawa Tengah's textile collapse was already building well before July 2024: API reported ~10,800 textile layoffs Jan-May 2024, with several named closures preceding July (PT Sinar Pantja Djaja ~2,000, PT Bitratex ~400, PT Djohartex ~300, PT Dupantex ~700).

Verdict: Elevated evenly across the year; the headline event is mis-dated. The single largest spike in the dataset rests on an attribution needing correction — the real discrete layoffs happened six months earlier.

Batch 4 — Scattered 2024 (Banten, DKI Jakarta, Jawa Timur, DIY)

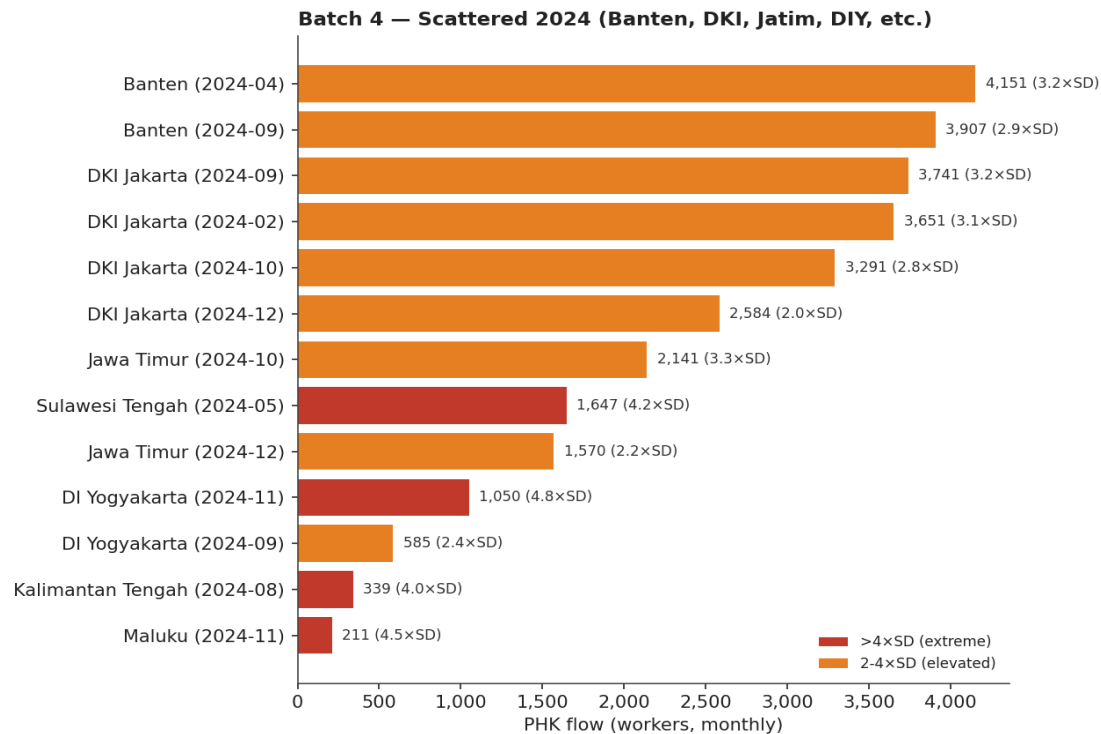


Figure 5. Batch 4 province-month spikes, sorted by magnitude.

Province	Month	PHK Flow	SD Multiple	Tier
Banten	2024-04	4,151	3.17×	2-4SD
Banten	2024-09	3,907	2.93×	2-4SD
DKI Jakarta	2024-09	3,741	3.21×	2-4SD
DKI Jakarta	2024-02	3,651	3.12×	2-4SD
DKI Jakarta	2024-10	3,291	2.75×	2-4SD
DKI Jakarta	2024-12	2,584	2.01×	2-4SD
Jawa Timur	2024-10	2,141	3.29×	2-4SD
Jawa Timur	2024-12	1,570	2.19×	2-4SD
Sulawesi Tengah	2024-05	1,647	4.17×	4SD+
DI Yogyakarta	2024-11	1,050	4.84×	4SD+
DI Yogyakarta	2024-09	585	2.41×	2-4SD
Kalimantan Tengah	2024-08	339	4.01×	4SD+
Maluku	2024-11	211	4.51×	4SD+

Original attribution (Round 1)

PT Sepatu Bata closure (233 workers, April 2024); PT Asia Pacific Fibers/POLY Karawang closure (2,500 workers, November 2024); PT Aditec/Quantum bankruptcy (511 workers, ~July 2024) for DKI Jakarta.

Pace verification (Round 2)

- Correction: PT Sepatu Bata's Purwakarta plant is in Jawa Barat, not Banten — an error in the original attribution. Bata had been loss-making since 2019 (net loss up 75.83% by 2023) and closing stores since 2021 — pre-existing distress, not a sudden April 2024 event.
- PT Asia Pacific Fibers (POLY): APSyFI's chairman stated the company “sudah melakukan PHK, terutama sejak tahun 2023” (already laying off, especially since 2023) — 1,200 workers already cut before the cited November 2024 closure; the “2,500/November 2024” figure compresses a multi-year staged process into a single date.
- National PHK rose steadily and evenly across 2024 (see Batch 3 finding) — meaning specific-month attributions for Banten/DKI/Jatim likely overstate how discrete these events actually were relative to the year's overall trend.

Verdict: Same pattern as Batch 3 — pre-existing distress, mostly lagged or relabeled reporting rather than fresh discrete shocks.

Batch 5 — July–November 2025 (Trump Tariff Period)

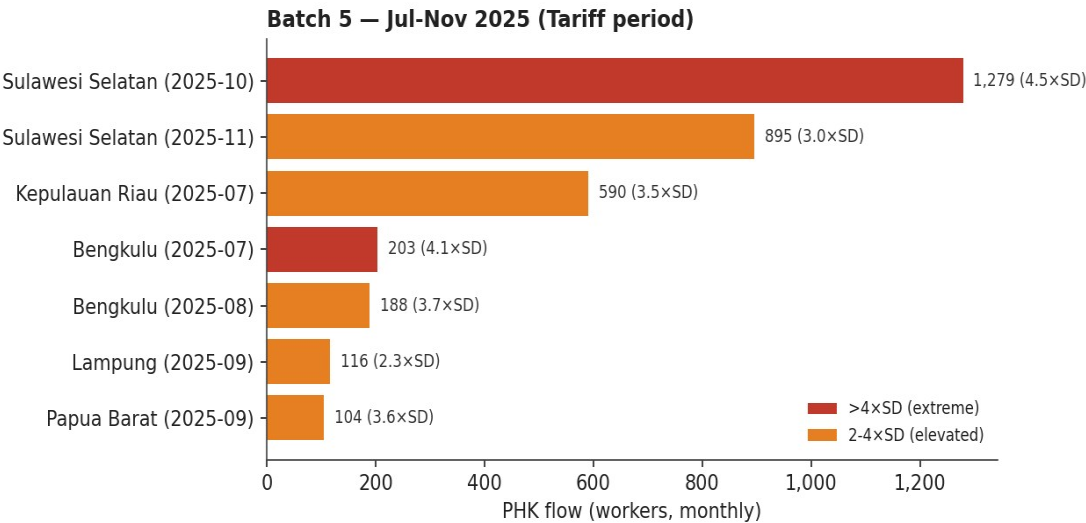


Figure 6. Batch 5 province-month spikes, sorted by magnitude.

Province	Month	PHK Flow	SD Multiple	Tier
Sulawesi Selatan	2025-10	1,279	4.50×	4SD+
Sulawesi Selatan	2025-11	895	2.95×	2-4SD
Kepulauan Riau	2025-07	590	3.48×	2-4SD
Sumatera Selatan	2026-01	467	3.19×	2-4SD
Bengkulu	2025-07	203	4.07×	4SD+
Bengkulu	2025-08	188	3.73×	2-4SD
Papua Barat	2025-09	104	3.64×	2-4SD

Original attribution (Round 1)

Trump's 32% reciprocal tariff (announced April 2025, effective 1 August 2025, later cut to 19%), with CELIOS estimating 1.2 million workers potentially at risk.

Pace verification (Round 2) — the most rigorously tested episode

- Monthly Kemnaker data shows layoffs peaked in January-February 2025 (17,796 in February alone) — months before the tariff was even formalized — driven by the Sritex bankruptcy (11,025 workers, following a Supreme Court ruling on 18 December 2024 and a 2021 debt default).

- The months immediately following tariff implementation (August 830, September 1,093, November 565) were the lowest of the entire year — the opposite of acceleration.
- The one prominent case where the tariff was directly named as a layoff cause (PT Victory Chingluh, ~2,200 workers, October 2025) was contested by its own union (KASBI), which cited product-quality/order-return issues instead.
- Full-year 2025 layoffs (88,519) did exceed 2024 (77,965), but this continues the same structural uptrend visible since 2022 (25,114 → 64,855 → 77,965 → 88,519), with a smaller percentage increase than prior years.

Verdict: No acceleration. The clearest confirmation of the “already at the cliff” pattern in the entire dataset.

3. Cross-Episode Synthesis

Episode	Cited Cause	Genuine Acceleration?	What the Pace Data Shows
Nov-Dec 2022	Textile export collapse + tech layoffs	No	Distress building since H2 2022; tech layoffs spread all year
Jun-Sep 2023	Coal price correction	No	Price decline gradual all year, steepest in May; true acceleration was Oct-Nov
Jul 2024	PT Sai Apparel bankruptcy	No — attribution mis-dated	Real layoffs (1,482) executed 31 Dec 2023; 2024 PHK rose evenly all year
Jul-Nov 2025	Trump 32% tariff	No — inverse pattern	Layoffs peaked Jan-Feb (Sritex), decelerated after tariff took effect

The consistent structural finding: across four independent episodes spanning three years, in every case the province-level PHK spikes in the data reflect the point at which a pre-existing, gradually building fragility became statistically visible — not the point at which a new, discrete shock began. This recurs whether the proposed cause is a global export shock, a commodity price cycle, a single company bankruptcy, or a trade-policy shock.

4. Implications for the ARIMA / Intervention Design

- Favor regime/trend variables over sharp single-month dummies — none of the four “causes” produced a clean discrete jump, so a single-month intervention dummy at the peak month will likely misattribute a slow-building trend to one point in time.
- One continuous “structural fragility” regime (2022 H2 onward) may fit better than four separate shock dummies, with named peak-events layered on top as secondary markers.
- Kalimantan Timur (Dec 2022) should be treated as a candidate data-quality issue, not modeled as an event — no corroborating narrative was found even after broadened searching.
- Flag known data-quality/sourcing conflicts explicitly in the methodology note: the 358,809 (2023) outlier vs. 64,855 official; the Sai Apparel 8,000/July-2024 vs. 1,482/Dec-2023 discrepancy; and the general Kemnaker undercount relative to Apindo/KSPN association tallies.

5. What Would Change These Conclusions

For all four episodes, the “accelerant, not shock” reading rests on cumulative/aggregate reporting rather than clean month-by-month provincial series. The most valuable next step: pull the raw month-by-month (not cumulative) `phk_flow` series directly from `phk_master.csv` for each flagged province, and test whether any single month's incremental change exceeds ~2 standard deviations above that province's trailing 6-month mean, independent of the cumulative national reporting used in this media research. If such a test showed a genuine one-month provincial outlier coinciding with a cited event, that episode could be reclassified as a discrete shock.

6. Master Source Caveats

- Kemnaker's official PHK figures systematically undercount true layoffs — they capture only JKP-eligible/reported cases, while Apindo, API, APSyFI, and KSPN consistently report substantially higher numbers from the same periods.
- Cumulative figures are retroactively revised for months after initial publication, as workers file JKP claims within a six-month window.
- Several headline figures across sources conflict and are flagged rather than silently resolved: the 358,809 (2023) outlier, the Sai Apparel dating conflict, and Kemnaker's 63,806 vs. 64,855 for 2023.
- Provincial monthly-level breakdowns for several 2024 cases were not directly obtainable at single-month granularity — the “evenly elevated” conclusion for 2024 rests on the national cumulative series plus each named firm's documented pre-existing distress.

- Some coal-sector news retrieved during research pertains to a 2026 licensing dispute unrelated to the 2022-2023 period and was explicitly excluded to avoid conflating separate events.